

WHATSAPP RAG CHATBOT SYSTEM DESIGN

This system connects a WhatsApp chatbot to a knowledge base built from uploaded documents.

It ensures that every user is first verified before accessing intelligent responses.

CORE FLOW:

A user sends a message → the system receives it via webhook → the system checks user state → if not verified, onboarding is triggered → if verified, the query is processed using RAG → response returned.

USER ONBOARDING:

New users must provide:

1. Name
2. Phone number

Only after successful verification, the system enables question answering.

RAG PIPELINE:

User query is converted into meaning (embedding), matched with stored knowledge, and then a language model generates a polished response.

ADMIN CONTROL:

The admin dashboard allows:

- Viewing logs (queries, responses, context)
- Triggering updates from document sources (Google Drive)

SYSTEM PRINCIPLES:

- Separation of ingestion and chat
- Single embedding model reused
- Vector database for similarity search
- Controlled access via user states

KNOWLEDGE BASE (EXAMPLE FILE REPRESENTATIONS)

The system processes multiple documents into structured knowledge chunks.

Below are 10 example text representations:

File 1: AI Overview

Artificial Intelligence focuses on enabling machines to simulate human intelligence, including reasoning, learning, and perception.

File 2: Cloud Computing

Cloud computing provides scalable resources over the internet, allowing flexible storage and computation.

File 3: Cybersecurity Basics

Cybersecurity protects systems and networks from digital attacks, ensuring confidentiality and integrity.

File 4: Data Structures

Efficient data organization like arrays, trees, and graphs is critical for performance.

File 5: Networking Concepts

Protocols like TCP/IP enable communication across distributed systems.

File 6: Machine Learning

Machine learning allows systems to learn patterns from data without explicit programming.

File 7: Database Systems

Databases manage structured data with consistency and efficient querying.

File 8: Web Development

Modern web apps use frontend-backend separation for scalability and maintainability.

File 9: Operating Systems

Operating systems manage hardware resources and provide user interfaces.

File 10: Software Engineering

Software engineering applies systematic approaches to design, development, and maintenance.

These files are chunked, embedded, and stored in a vector database for retrieval during user queries.